LESSON 1.2

Math is Exploring and Thinking

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make sense of a problem, plan a strategy, and use fractions of a whole number to compare quantities.

WHY THIS MATTERS Doing math is exploring: you make sense of a problem, plan a strategy, check your progress, and try new paths when you hit a dead end. Today's problems — walking distances and skyscraper heights — reward thinkers, not just calculators. Try this one: a new building will be 3/4 as tall as a tower that is 400 meters tall. Plan a strategy to find how tall the new building will be.

REMEMBER

Math is Exploring and Thinking

Math Practices: Sense Making & Strategy Planning The book opens with two warm-ups: about how many peas are in the picture, and decompose 105.76 five different ways. Both are about thinking before computing.

1. Identify known quantities and the unknown question.
2. Choose a strategy (draw a model, write an equation, or build a table).
3. Evaluate progress and pivot to an alternate approach if needed

Words: quantity *(cantidad)* · relationship *(relación)* · strategy *(estrategia)* · reasonable *(razonable)* · persevere *(perseverar)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each building plan: shorter or taller than 540 meters?

Groups: Shorter than 540 m (fraction less than 1) · Taller than 540 m (fraction greater than 1)

_____ 9/10 of 540

_____ 5/6 of 540

_____ 3/4 of 540

_____ 6/5 of 540

_____ 11/10 of 540

_____ 7/6 of 540

HINT Compare the numerator to the denominator before you multiply anything.

2 **Complete the table. Show how you found each value.**

Compute each building's height as a fraction of 540 meters.

Building plan	Fraction of 540 m	Height (m)
Rio de Janeiro		
Tokyo		
Prototype		

HINT Divide 540 by the denominator first, then multiply by the numerator.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A Ferris wheel has 10 cars. Each car holds 4 people. How many people ride at once?

(A) 14 people
(B) 20 people
(C) 40 people
(D) 44 people

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is half of 20?

(A) 5
(B) 10
(C) 15
(D) 40

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A pizza is cut into 8 equal slices. You eat half of it. How many slices did you eat?

(A) 2 slices
(B) 4 slices
(C) 8 slices
(D) 16 slices

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Pick one card you put under 'We DON'T know this yet.' What would you have to figure out first to move it to the known column, and which quantity does it depend on?

SENTENCE STARTER To know ___, I first need ___, because ___ depends on it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 1.2

Math is Exploring and Thinking

Set B · Version B · Second form · core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make sense of a problem, plan a strategy, and use fractions of a whole number to compare quantities.

WHY THIS MATTERS Doing math is exploring: you make sense of a problem, plan a strategy, check your progress, and try new paths when you hit a dead end. Today's problems — walking distances and skyscraper heights — reward thinkers, not just calculators. Try this one: a new building will be 3/4 as tall as a tower that is 400 meters tall. Plan a strategy to find how tall the new building will be.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Deon walked a shorter distance than Miguel, and Evelyn walked a longer distance than Miguel. You never learn the exact distances — but the relationships still tell you a lot. Building height problem: which statement shows you are CHECKING your solution against the relationship in the problem?

(A) “5/6 is less than 1, so my answer for London should be less than 540 meters”

(B) “My answer is 650 meters, and that seems like a big number so it's probably right”

(C) “I don't need to check once I've multiplied”

(D) “Since I used a fraction, the answer must be exact”

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which statement best shows what this lesson means by "math is exploring and thinking"?

(A) We make sense of problems, plan a strategy, check our progress, and try other strategies at dead ends

(B) We compute as fast as possible and move on to the next problem

(C) We wait for someone to show us the one correct method

(D) We only check our work when the answer looks wrong

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Deon walked a shorter distance than Miguel, and Evelyn walked a longer distance than Miguel. You never learn the exact distances — but the relationships still tell you a lot. In the Comparing Walks problem, Deon walked a shorter distance than Miguel, and Evelyn walked a longer distance than Miguel. Without any exact distances, what can you determine?

(A) Nothing — you need exact numbers before you can compare

(B) Deon and Evelyn walked the same distance

(C) The order from shortest to longest: Deon, Miguel, Evelyn

(D) Evelyn walked the shortest distance

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Deon walked a shorter distance than Miguel, and Evelyn walked a longer distance than Miguel. You never learn the exact distances — but the relationships still tell you a lot. A student picks Miguel = 2 miles, Deon = 2.5 miles, Evelyn = 1.5 miles. What is wrong with these values?

(A) Nothing is wrong — any three distances work

(B) Evelyn's distance is too small compared to Deon's

(C) Deon's distance is greater than Miguel's, which breaks “shorter than”

(D) Miguel's distance must always be the smallest of the three

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Pick one card you put under 'We DON'T know this yet.' What would you have to figure out first to move it to the known column, and which quantity does it depend on?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help